

MAT3280 Midterm exam – sample solution

10/28/2023

Question 1. (10 points) Suppose we have a mechanism to generate a random variable U that is uniformly distributed between 0 and 1. Find a function h such that $X = h(U)$ is exponentially distributed with mean $1/\lambda$, where λ is a fixed positive constant.

The probability density function of exponential random variable is

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x} & \text{for } x \geq 0, \\ 0 & \text{for } x < 0. \end{cases}$$

Solution. The cumulative distribution function of an exponentially distributed random variable is

$$F_X(x) = \begin{cases} 1 - e^{-\lambda x} & \text{for } x \geq 0 \\ 0 & \text{otherwise.} \end{cases}$$

We solve for the inverse function of $F_X(x)$ by solving

$$y = F_X(x).$$

The inverse function is

$$h(y) = -\frac{1}{\lambda} \ln(1 - y),$$

where y is a variable between 0 and 1. We can take the function h and define random variable X by

$$X = -\frac{1}{\lambda} \ln(1 - U)$$

The distribution of X will be exponential with mean $1/\lambda$. To see why X has the desired distribution, we can take any positive real number x , and consider the probability

$$\begin{aligned} \Pr(X \leq x) &= \Pr\left(-\frac{1}{\lambda} \ln(1 - U) \leq x\right) \\ &= \Pr(U \leq 1 - e^{-\lambda x}). \end{aligned}$$

Because U is uniformly distributed, the last probability is equal to $1 - e^{-\lambda x}$. Therefore

$$\Pr(X \leq x) = 1 - e^{-\lambda x}.$$

This proves that the cdf of X is the same as the cdf of the exponential distribution with mean $1/\lambda$.

(Alternate solution: Since U and $1 - U$ has the same distribution, we can simplify a little bit and take $h_1(u) = -\lambda^{-1} \ln(u)$. The random variable $h_1(U)$ has the same distribution as $h(U)$.)

Question 2. (10 points) Let $G(x)$ be the function

$$G(x) \triangleq \begin{cases} 1 - \frac{1}{x^2+2} & \text{for } x > 0, \\ 0 & \text{for } x \leq 0. \end{cases}$$

Is it possible to construct a real-valued random variable X such that the probability $P(X \leq x)$ is equal to $G(x)$, for all real numbers x ?

Solution. The function $G(x)$ is not a Stieltjes measure function, because at the point $x = 0$, the function $G(x)$ is continuous from the left, not continuous from the right. Based on this observation, the answer to this question is no.

In more detail, suppose such a random variable X exists. This means that there is a probability space $(\Omega, \mathcal{F}, P)$, and $X(\omega)$ is a measurable function such that for all real numbers x , the event

$$\{\omega \in \Omega : X(\omega) \leq x\}$$

has probability $G(x)$. We would have $P(X \leq 0) = G(0) = 0$. However, if we consider the events

$$A_n \triangleq \{\omega \in \Omega : X(\omega) \leq 1/n\}$$

for positive integer n , we see that the events A_n form a decreasing sequence of sets, with intersection

$$B \triangleq \{\omega \in \Omega : X(\omega) \leq 0\}$$

By upper semi-continuity of measure, which should hold in general, we would get

$$\lim_{n \rightarrow \infty} P(A_n) = P(B) = G(0) = 0.$$

However, the numerical value of the limit on the left is

$$\lim_{n \rightarrow \infty} P(A_n) = \lim_{n \rightarrow \infty} 1 - \frac{1}{(1/n)^2 + 2} = \frac{1}{2}.$$

This is a contradiction and hence no such random variable X exists.

Question 3. (16 points) Let $\mathcal{F}$ denote the σ -algebra generated by closed intervals of the form $[a, b]$, where $a \leq b$ are rational numbers. Show that the following sets are $\mathcal{F}$ -measurable.

- (a) The set of all integers.
- (b) Open interval (x_1, x_2) , where x_1 and x_2 are real numbers with $x_1 < x_2$.

Solution.

(a) If we set $a = b$, the closed interval $[a, b]$ reduces to a single point. By taking a and b to be equal to an integer n , which is certainly a rational number, we see that the set $\{n\}$ consisting of

integer n is $\mathcal{F}$ -measurable. Since the set of integers are countable, we can write it as a countable union

$$\mathbb{Z} = \bigcup_{n \in \mathbb{Z}} \{n\}.$$

Since $\mathcal{F}$ is supposed to be closed under countable union, we conclude that the set of all integers is $\mathcal{F}$ -measurable.

(b) We take a sequence of rational numbers a_n , that are all strictly larger than x_1 and converging to x_1 from the right, and take another sequence of rational numbers b_n , that are all strictly less than x_2 and converging to x_2 from the left.

The reason why we can pick these two sequences is because rational numbers are dense in the real number line. We can start with choosing a_1 to be any rational number that is strictly larger than x_1 . Take the mid-point $(x_1 + a_1)/2$. Because we want the value of a_2 to be strictly larger than x_1 as well, we pick a sufficiently small real number ϵ so that the interval

$$[x_1 + \epsilon, (x_1 + a_1)/2]$$

is nonempty. By the property that rational numbers are dense, we can find a rational number a_2 that is within the above interval. Recursively, after a_k is chosen, we pick a_{k+1} to be a rational number that is strictly between x_1 and $(x_1 + a_k)/2$. Throughout this process, the value of a_n is strictly larger than x_1 for all n , and a_n is a decreasing sequence converging to x_1 from the right.

Similar argument guarantee that we can choose the sequence b_n 's with the required properties.

Then, we can write the open interval (x_1, x_2) as a countable union

$$(x_1, x_2) = \bigcup_{n=1}^{\infty} [a_n, b_n].$$

Because $[a_n, b_n]$ are $\mathcal{F}$ -measurable, the union (x_1, x_2) is also $\mathcal{F}$ -measurable.

We note that when we take the union, the two boundary point x_1 and x_2 are not included.

Question 4. (16 points)

- (a) Prove that if $\mathcal{F}_1$ and $\mathcal{F}_2$ are σ -algebras on a common sample space Ω , then the intersection $\mathcal{F}_1 \cap \mathcal{F}_2$ is also a σ -algebra. (The notation $\mathcal{F}_1 \cap \mathcal{F}_2$ means the collection consisting of sets that are member of both $\mathcal{F}_1$ and $\mathcal{F}_2$.)
- (b) Give an example of two σ -algebras $\mathcal{F}_1$ and $\mathcal{F}_2$ on a common sample space Ω such that the union $\mathcal{F}_1 \cup \mathcal{F}_2$ is not a σ -algebra.

Solution. (a) We assume that $\mathcal{F}_1$ and $\mathcal{F}_2$ satisfy the axioms of σ -algebra, and show that $\mathcal{F}_1 \cap \mathcal{F}_2$ also satisfies the axioms.

- Because $\mathcal{F}_1$ and $\mathcal{F}_2$ are both σ -algebra, Ω belongs to both $\mathcal{F}_1$ and $\mathcal{F}_2$. Therefore Ω is a set in $\mathcal{F}_1 \cap \mathcal{F}_2$.
- Suppose A is a set in $\mathcal{F}_1 \cap \mathcal{F}_2$. Because A is an element in $\mathcal{F}_1$, A^c is an element in $\mathcal{F}_1$. On the other hand, because A is in $\mathcal{F}_2$, A^c is in $\mathcal{F}_2$. Therefore A^c is in $\mathcal{F}_1 \cap \mathcal{F}_2$.

- Suppose $(A_n)_{n=1}^\infty$ is a sequence of sets in $\mathcal{F}_1 \cap \mathcal{F}_2$. For $i = 1, 2$, we have $A_n \in \mathcal{F}_i$ for all n . Since $\mathcal{F}_i$ is a σ -algebra, we have $\bigcup_{n=1}^\infty A_n \in \mathcal{F}_i$. Because it holds for $i = 1$ and $i = 2$, we obtain

$$\bigcup_{n=1}^\infty A_n \in \mathcal{F}_1 \cap \mathcal{F}_2.$$

(b) We construct a finite example with $\Omega = \{a, b, c\}$. The first σ -algebra $\mathcal{F}_1$ is

$$\mathcal{F}_1 \triangleq \{\emptyset, \{a\}, \{b, c\}, \Omega\},$$

and the second σ -algebra $\mathcal{F}_2$ is

$$\mathcal{F}_2 \triangleq \{\emptyset, \{c\}, \{a, b\}, \Omega\}.$$

The union $\mathcal{F}_1 \cup \mathcal{F}_2$ is the following collection of sets,

$$\mathcal{F}_1 \cup \mathcal{F}_2 = \{\emptyset, \{a\}, \{c\}, \{a, b\}, \{b, c\}, \Omega\}.$$

This collection of sets fails to be a σ -algebra, because it is not closed under union. The union of $\{a\}$ and $\{c\}$ is not a member of $\mathcal{F}_1 \cup \mathcal{F}_2$.

(An alternate argument is to use the fact that a σ -algebra over a finite sample space Ω must contain 2^k subsets, for some integer k . The union $\mathcal{F}_1 \cup \mathcal{F}_2$ contains 6 subsets, and 6 is not a power of 2, hence it cannot be a σ -algebra.)

Question 5. (24 points) John plays a game an infinitely number of times. The game is designed in such a way that the outcomes at different times are independent, and the probability of winning the n -th game is $1/n$, where $n = 1, 2, 3, \dots$. Hence, if John plays this game, he will win the first game with probability 1, the second game with probability $1/2$, and the third game with probability $1/3$, and so on.

- For $n \geq 1$, let E_n be the event that John win at both time n and time $n + 1$. Are the sequence of events $(E_n)_{n=1}^\infty$ independent?
- Determine the probability that John wins an infinitely number of times.
- Find the probability that John wins in two consecutive games an infinite number of times.

Solution. (a) The events E_n 's are not independent. Consider the events E_2 and E_3 . The probabilities of events E_2 and E_3 are

$$P(E_2) = \frac{1}{2} \cdot \frac{1}{3} = \frac{1}{6}$$

$$P(E_3) = \frac{1}{3} \cdot \frac{1}{4} = \frac{1}{12}$$

The union $E_2 \cap E_3$ is the event that John wins at time 2, 3 and 4. This happens with probability

$$P(E_2 \cap E_3) = \frac{1}{2} \cdot \frac{1}{3} \cdot \frac{1}{4} = \frac{1}{24}.$$

It is clear that

$$P(E_2 \cap E_3) \neq P(E_2) \cdot P(E_3).$$

Therefore E_2 and E_3 are dependent events. If any two events in the sequence are dependent, the whole sequence of events is dependent.

(b) We check that the infinite series

$$\sum_{n=1}^{\infty} P(\text{John wins at the } n\text{-th game}) = \sum_{n=1}^{\infty} \frac{1}{n} = \infty$$

is divergent. By the second Borel-Cantelli lemma, John wins an infinitely number of times with probability 1.

(c) The events E_n 's are dependent, but in the first Borel-Cantelli lemma, it is not required that the events are independent. The sum of probabilities

$$\sum_{n=1}^{\infty} P(E_n) = \sum_{n=1}^{\infty} \frac{1}{n(n+1)} < \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

is finite. Therefore, John wins in two consecutive games infinitely often with probability 0.

(We show in part (b) that

$$P(\text{infinitely many wins}) = 1,$$

but in part (c) we see that

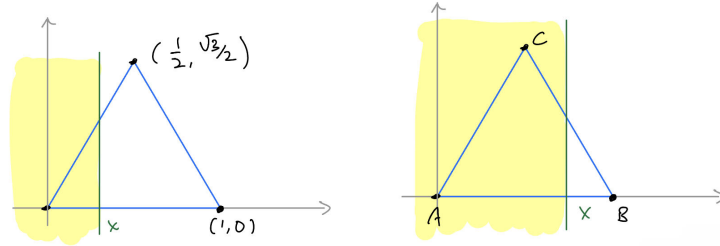
$$P(\text{infinitely many consecutive wins}) = 0.$$

This is possible. We can also check that, with probability 1, almost all winning game will follow by a losing game.)

Question 6. (24 points) Let T represent the boundary of equilateral triangle with vertices $A = (0, 0)$, $B = (1, 0)$ and $C = (1/2, \sqrt{3}/2)$. (The interior of the triangle is not included in T .) We generate a random point (X, Y) uniformly on T . This means that, with probability $1/3$, the random point (X, Y) is uniformly chosen on the line segment AB , and with probability $1/3$, (X, Y) is uniformly chosen on BC , and with probability $1/3$, (X, Y) is uniformly chosen on CA .

- (a) Describe the distribution of the x -coordinate X of the random point.
- (b) Describe the distribution of the y -coordinate Y of the random point.
- (c) Compute the expectation of X and the expectation of Y .

Solution. (a) From the geometry, we can see that the distribution of X should be uniform between 0 and 1.



Let's verify it formally by computing $P(X \leq x)$. The probability $P(X \leq x)$ is the length of the portion of the triangle in the shaded area, divided by the total length 3. When $0 \leq x \leq 1/2$, the region to the left of vertical line with x -coordinate x is shown in the above figure. It includes the portion of the horizontal sides from $(0,0)$ to $(x,0)$, and the slanted side from $(0,0)$ to $(x, \sqrt{3}x)$. The probability that a random point on T will fall on this region is

$$P(X \leq x) = \frac{1}{3} \left(x + \frac{x}{\cos(\pi/3)} \right) = x.$$

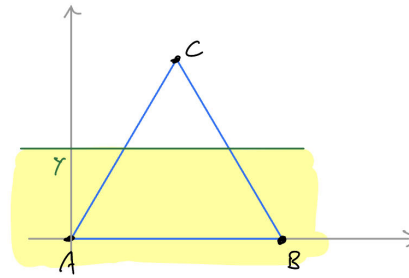
When $1/2 \leq x \leq 1$. The region to the left of the vertical line with x -coordinate x include the whole line segment. In this case, we have

$$P(X \leq x) = \frac{1}{3} \left(1 + x + \frac{x - 1/2}{\cos(\pi/3)} \right) = x.$$

Hence, for $0 \leq x \leq 1$, $P(X \leq x) = x$. The random variable X is thus uniformly distributed between 0 and 1.

(b) The cdf of Y has a discontinuity at the point $y = 0$. When $y < 0$, we certainly has $P(Y \leq y) = 0$. However, the region $Y \leq y$ include the line segment AB . As a result, we have

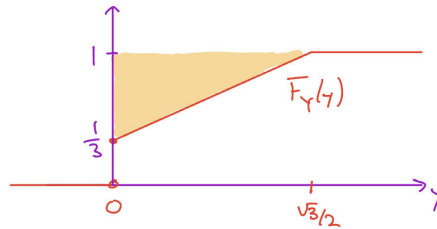
$$P(Y \leq 0) = 1/3$$



When $0 < y \leq \sqrt{3}/2$, the portion of the triangle that is in the above shaded area consists of three parts: the whole line segment AB , part of AC and part of BC . The probability that a random point chosen from the triangle will be in the shaded region is

$$P(Y \leq y) = \frac{1}{3} \left(1 + 2 \frac{y}{\sin(\pi/3)} \right) = \frac{1}{3} \left(1 + \frac{4y}{\sqrt{3}} \right)$$

The cumulative distribution function of Y is shown in the graph below



(c) Because X is uniformly distributed, the expectation of X can be computed easily and is equal to $1/2$.

For nonnegative random variable, we can use the formula

$$E[Y] = \int 1 - F_Y(y) dy$$

to compute the expectation of Y . This is the area of the shaded region in orange color in the above graph. The expectation of Y is

$$E[Y] = \frac{\frac{2}{3} \cdot \frac{\sqrt{3}}{2}}{2} = \frac{\sqrt{3}}{6}.$$